

Efficacy trial on the purified compounds of the seeds of *Carica papaya* for male contraception in albino rat

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Abstract

The contraceptive efficacy and toxicological screening of the two principal compounds, MCP I and ECP I, isolated from the seeds of *Carica papaya*, in male albino rats at the standardized dose regimen, at 50 mg/kg b.w./day, for a period of 360 days and up to 90 days of treatment withdrawal have been reported. The body and organ weights, cauda epididymal sperm characteristics, androgen sensitive tissue biochemistry, reactive oxygen species and anti-oxidant defense system in the cauda epididymal microenvironment, histology and ultrastructure of testis and cauda epididymis, histology of seminal vesicle and prostate, toxicological investigations through routine hematology and serum clinical chemistry, sexual behaviour and fertility index have been studied. The results revealed that oral administration of MCP I and ECP I were equally effective, exhibiting complete inhibition of sperm motility following 90 days of treatment that coincided with a gradual and significant decline in cauda epididymal sperm density, percent viable spermatozoa and significant increase in sperm anomalies. Histology of testis of treated animals revealed degenerated germinal epithelium, vacuolization in Sertoli cells and proliferating germ cells and disturbances in spermatid differentiation. Spermatogonial stem cell reserves and Leydig cells appeared normal. Ultrastructure of the testis revealed vacuolization in the Sertoli cells and germ cells, loss of cytoplasmic characteristics in the Sertoli cells, nuclear degeneration and mitochondrial vacuolization in spermatocytes and spermatids. Leydig cells exhibited steroidogenic features. Cauda epididymis showed normal epithelial cell function. Absence of spermatozoa or disruption of spermatozoa clusters in the lumen were evident. Ultrastructure of cauda epididymis revealed normal secretory activity. Morphology of seminal vesicle and prostate of the treated animals were comparable to control animals. Serum testosterone, tissue biochemical and toxicological parameters remained unaffected. Fertility test revealed 100% efficacy. All the altered parameters showed sign of recovery following 90 days of treatment withdrawal. It is concluded that both MCP I and ECP I are equally effective in terms of contraceptive efficacy which is likely reversible and without adverse side effects.

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1. Introduction

Recent efforts are being devoted to identify a plant-based male contraceptive, one that is supposedly orally bioactive, non-toxic and more importantly cost effective, based on ethnomedical information [1]. Among plant products so far tested for male anti-fertility *Gossypol*, a yellow phenolic

compound isolated from the cotton seed oil, and *Triptolide*, a diterpene triepoxide from *Tripterygium wilfordii* have shown promising results in animal models and humans; however, undesirable side effects such as delayed reversibility or irreversibility and hypokalemic paralysis of the former [2], and the effect on T cell proliferation and delayed reversibility of the later [3–4], raised serious concerns about their clinical application in male contraception.

Seeds of *Carica papaya* have shown promise in male contraception. Preliminary findings in rats and rabbits given different extracts/fractions of the seeds with variable dose, duration and mode of administration, revealed maximum

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anti-fertility in the chloroform extract [5–6]. This contraceptive effect was attributed to mild estrogenicity [7]. Efforts to further purify the bioactivity revealed activity in the benzene chromatographic fraction of the chloroform extract that inhibited cauda epididymal sperm motility, reduced sperm density, and altered sperm morphology in rats and/or rabbits [8–9]. The methanol and ethyl acetate sub-fractions of the benzene chromatographic fraction also showed sperm motility inhibition in rats following 30–60 days of treatment and azoospermia in rabbits within 15 days of treatment [1,9]. Although general lack of observable toxicity, and the full recovery following cessation of treatment were reported in these studies, the mode of action of the bioactivity may differ between rats and rabbits and thus be species-specific.

Continued efforts to further purify the benzene fraction with thin-layer chromatography (TLC) yielded two principal components each for the methanol subfraction (MCP I, II) and the ethyl acetate subfraction (ECP I, II). In the present study, the major components of these subfractions, namely MCP I and ECP I, were further investigated in rats for efficacy in male contraception, for general toxicity, and for further insights into the mode of action.

2. Materials and methods

2.1. Drug preparation

Fresh seeds of *C. papaya* L. (Caricaceae, Angiosperm, Voucher No. RUBL 16590) of the honey dew variety were obtained commercially and shade dried. Dried seeds (600 g) were coarsely powdered and refluxed with 3 L of chloroform at 58° C for 12 × 3 h. The extract was concentrated under reduced pressure and dried. The residue received from the chloroform extract was chromatographed over deactivated silica gel (60–120 mesh) in a glass column. The different fractions were eluted sequentially with 1 L of solvents of increasing polarity. The benzene fraction, second in the order and the most effective in preliminary bioassays, was concentrated under reduced pressure and fractionated with ethyl acetate and methanol, 300 ml each. The methanol and ethyl acetate sub-fractions were subjected to TLC and eluted with petroleum spirit, benzene, ethanol and double distilled water (5:5:1:1). Two major compounds were isolated from the methanol and ethyl acetate subfractions, designated MCP I and II and ECP I and II, respectively. Preparative scale amounts of these compounds were obtained by repeating the TLC procedures.

2.2. Animals

Clinically healthy, proven fertile, colony bred Sprague–Dawley rats (*Rattus norvegicus*), approximately 3 months old, weighing 180–200 g were used in the present research. The animals were housed three per polypropylene cage on a 12-h light:12-h dark cycle, and fed rat pellet diet (Ashirwad Industries, Chandigarh, India), soaked wheat, gram and

maize and water ad libitum. Animal procedures were approved by the Institutional Ethical Committee and conducted in compliance with the Guidelines for Care and Use of Animals for Scientific Research (Indian National Science Academy, 2000).

2.3. Experimental design

The experiments consisted of the following three groups of 45 rats per group. All animals were treated orally with the vehicle, 0.5 ml distilled water, given daily for 360 days. Group I (control) received the vehicle only. Group II (MCP I) received MCP I emulsified in distilled water at a dosage of 50 mg/kg b.w./day, and Group III (ECP I) received ECP I emulsified in distilled water at 50 mg/kg b.w./day. Due care was taken to assure that the animals received the complete dose. Three animals from each group were sacrificed every 30 days across the 360-day test period, and until 90-day treatment withdrawal, after recording the body weight (b.w.) and after completing the fertility test.

2.3.1. Body weight and organ weight

Body weight and the weight of the brain, thyroid, heart, lung, liver, spleen, adrenal, kidney, testis, epididymis, seminal vesicle and ventral prostate, were recorded in every autopsy schedule.

2.3.2. Sperm parameters

Cauda epididymal spermatozoa obtained by microepididymal puncture in normal saline were used for the analyses of sperm concentration, motility viability and abnormality according to WHO manual method [16].

2.3.3. Histology and Ultrastructure of reproductive organs

The testis, cauda epididymis, seminal vesicle and prostate were processed for histological and ultrastructural evaluation. For histology the tissues were fixed in Bouin's fluid, dehydrated and embedded in paraffin for section at 5 µm. These sections were stained with Harris hematoxylin and eosin. For ultrastructural studies, the tissues were fixed immediately in 2.5% glutaraldehyde in 0.1 M phosphate buffer, pH 7.2, for 24 h and then washed thoroughly in phosphate buffer, postfixed in 1% osmium tetroxide for 6 h, washed in phosphate buffer, dehydrated in acetone, and embedded in araldite 512 (60° C, 48 h). Thick sections (1 µm) were stained with toluidine blue and ultrathin sections were stained with uranyl acetate and lead citrate for observation under a transmission electron microscope (CM-10, Philips, Eindhoven, The Netherlands).

2.3.4. Tissue biochemistry

Assays were conducted to determine levels of metabolites and enzymatic activities as follows: cholesterol [10], glycogen [11] and lactate dehydrogenase (LDH) [12] in the testis; sialic acid [13], L-carnitine [14] and α-glucosidase [15] in the epididymis; fructose and acid phosphatase (ACP) in the

seminal vesicle and ventral prostate, respectively [15]. One side of the cauda epididymis was also used to assay for lipid peroxidation (LPO) [17–18], catalase (CAT) [19], superoxide dismutase (SOD) [20], and reduced glutathione (GSH) [21].

2.4. Toxicology and fertility assessment

2.4.1. Hematology

Blood samples were collected by cardiac puncture and used for total red blood cells (RBC), white blood cells (WBC) [22], hemoglobin [23] and red cell indices such as packed-cell volume (PCV), mean corpuscular volume (MCV), mean corpuscular hemoglobin (MCH) and mean corpuscular hemoglobin concentration (MCHC) [24].

2.4.2. Clinical biochemistry

Serum protein, glucose, cholesterol, creatinine, serum glutamate pyruvate transaminase (SGPT), serum glutamate oxalate transaminase (SGOT), lactate dehydrogenase, creatine kinase (CK), bilirubin, urea, triglycerides (TGL) and high density lipoprotein (HDL) were recorded using reagent kits (Transasia Biomedicals Ltd., Mumbai, India). Serum testosterone was estimated by enzyme immunoassay (Biochem Immuno Systems, Italy).

2.4.3. Libido

Libido was assessed by observing mounting behaviors. Control or treated males were paired in a 1:2 ratio with control, proven fertile females in proestrus. The pre-coital sexual behaviors such as chasing, nosing, anogenital sniffing and mounting were observed for up to 2 h of pairing.

2.4.4. Fertility test

Rats were subjected to fertility test, at monthly intervals, with proven fertile females (1:2). Success of mating was confirmed in vaginal smear. The females were allowed to complete gestation to term and offspring records were maintained.

2.4.5. Statistical analysis

Student's *t*-test was employed for statistical comparison.

3. Results

3.1. Body and organ weights

Daily (360 days) exposure to MCP I or ECP I led to a significant decrease in total body weight and the weights of several organs (testis, epididymis, seminal vesicle and prostate) relative to the vehicle control group (Tables 1A and 1B). Following withdrawal of MCP I or ECP I treatments there were signs of weight recovery in some organs at 90 days although the extent of recovery was far from complete. Treatment did not result in significant reductions in the weight of other organs, such as the brain, thyroid, heart, lungs, liver, spleen, adrenal and kidney (Tables 1A and 1B).

3.2. Sperm analysis

Sperm characteristics in MCP I or ECP I treated animals showed a significant reduction in all the parameters studied (motility, viability, density and percent normal sperm) from 30- to 360-day during treatment (Tables 2A and 2B). Total inhibition of sperm motility was observed uniformly after 90 days of treatment and this effect lasted through the full 360-day study period. There was gradual improvement in all sperm parameters monitored here following treatment withdrawal. Some parameters reached the control value 90 days of treatment withdrawal; however, in ECP I treated animals, the values showed significant reductions in motility and viability, and significant increases in sperm abnormality, even after 90 days of treatment withdrawal (Tables 2A and 2B).

3.3. Tissue biochemistry

Testes of exposed animals showed no appreciable alterations in cholesterol, glycogen or LDH levels. Epididymides of exposed animals showed normal sialic acid, L-carnitine and α -glucosidase levels. Fructose of seminal vesicles, and acid phosphatase of the prostate, showed minor fluctuations but no significant alterations for the MCP I and ECP I treatment groups (Tables 2A and 2B). No significant alterations were detected for lipid peroxidation, catalase, superoxide dismutase, and reduced glutathione of the epididymal fluid (data not shown).

3.4. Toxicological investigations

Clinical chemistry parameters (total RBCs, WBCs, Hb, PCV, MCV, MCH and MCHC and serum protein, glucose, cholesterol, creatinine, SGPT, SGOT, LDH, CK, bilirubin, urea, triglycerides and HDL) were not significantly different for the MCP I and ECP I treatments versus control animals throughout the treatment period or after withdrawal of treatment. Serum testosterone fluctuated within the control range in both treatment groups (data not shown).

3.5. Morphology of reproductive organs

3.5.1. Testis

Disruption of spermatogenesis was evident in the testis of treated animals after the 360-day exposure period. Vehicle control rats showed round or oval seminiferous tubules with normal Sertoli cells and germ cells at various stages covering the complete spermatogenesis, and an interstitium containing distinct Leydig cells (Fig. 1). Degeneration of the seminiferous epithelium was common in a majority of the seminiferous tubules examined from treated animals (Figs. 2–3). These Sertoli cells displayed vacuolization and germ cells with nuclear pyknosis and cytoplasmic vacuolization. Alterations appeared greatest during spermatid differentiation.

Ultrastructurally, the control testis showed features typical of normal spermatogenesis. Sertoli cells displayed

Table 1A

The body weight and organs weight in rats treated orally with the isolated compound MCP I of the methanol sub-fraction of the benzene chromatographic fraction of the chloroform extract of the seeds of *Carica papaya* at 50 mg/kg b.w./day (values are S.E.M. of three animals per group)

Treatment schedule	Body weight (g)	Brain (g)	Thyroid (g)	Heart (g)	Lungs (g)	Liver (g)	Spleen (g)
Treatment							
60 days	175 ± 1.45 (178 ± 1.21)	1.63 ± 0.13 (1.65 ± 0.08)	0.03 ± 0.01 (0.02 ± 0.01)	0.58 ± 0.09 (0.61 ± 0.21)	1.24 ± 0.05 (1.33 ± 0.42)	3.50 ± 0.14 (3.61 ± 0.42)	0.46 ± 0.06 (0.52 ± 0.08)
120 days	173 ± 2.50 (178 ± 1.20)	1.59 ± 0.20 (1.62 ± 0.42)	0.02 ± 0.01 (0.03 ± 0.01)	0.60 ± 0.08 (0.64 ± 0.20)	1.23 ± 0.33 (1.28 ± 0.56)	3.60 ± 0.47 (3.66 ± 0.80)	0.51 ± 0.09 (0.54 ± 0.12)
180 days	171 ± 3.21 (178 ± 2.12)	1.67 ± 0.34 (1.65 ± 0.82)	0.03 ± 0.01 (0.02 ± 0.01)	0.59 ± 0.04 (0.62 ± 0.06)	1.26 ± 0.14 (1.30 ± 1.00)	3.50 ± 0.41 (3.60 ± 0.48)	0.40 ± 0.13 (0.54 ± 0.18)
240 days	170 ± 1.54** (183 ± 2.00)	1.43 ± 0.31 (1.71 ± 0.96)	0.02 ± 0.01 (0.03 ± 0.02)	0.64 ± 0.07 (0.68 ± 0.08)	1.24 ± 0.31 (1.32 ± 0.55)	3.57 ± 0.06 (3.65 ± 0.81)	0.52 ± 0.03 (0.55 ± 0.18)
300 days	169 ± 2.25** (188 ± 1.32)	1.61 ± 0.45 (1.68 ± 0.62)	0.03 ± 0.01 (0.02 ± 0.01)	0.54 ± 0.16 (0.66 ± 0.32)	1.24 ± 0.35 (1.40 ± 0.88)	3.52 ± 0.23 (3.79 ± 0.80)	0.51 ± 0.06 (0.58 ± 0.21)
360 days	168 ± 1.29** (190 ± 2.24)	1.54 ± 0.43 (1.66 ± 0.84)	0.03 ± 0.02 (0.02 ± 0.01)	0.61 ± 0.08 (0.66 ± 0.18)	1.20 ± 0.34 (1.38 ± 0.62)	3.53 ± 0.31 (3.75 ± 0.96)	0.48 ± 0.10 (0.61 ± 0.81)
Treatment withdrawal							
30 days	172 ± 11.60* (213 ± 3.36)	1.58 ± 0.06 (1.66 ± 0.40)	0.04 ± 0.02 (0.03 ± 0.01)	0.64 ± 0.15 (0.70 ± 0.92)	1.29 ± 0.41 (1.44 ± 0.96)	3.50 ± 0.04 (3.81 ± 0.88)	0.42 ± 0.13 (0.60 ± 0.21)
90 days	171 ± 3.15*** (225 ± 2.84)	1.66 ± 0.24 (1.69 ± 0.36)	0.03 ± 0.01 (0.03 ± 0.02)	0.60 ± 0.06 (0.71 ± 0.38)	1.26 ± 0.16 (1.38 ± 0.78)	3.56 ± 0.34 (3.78 ± 0.64)	0.47 ± 0.01 (0.60 ± 0.12)
Treatment schedule	Adrenal (g)	Kidney (g)	Testis (mg/100 g b.w.)	Epididymis (mg/100 g b.w.)	Seminal Vesicle (mg/100 g b.w.)	Ventral prostate (mg/100 g b.w.)	
Treatment							
60 days	0.13 ± 0.01 (0.12 ± 0.02)	1.39 ± 0.10 (1.42 ± 0.16)	560 ± 1.76* (571 ± 2.00)	168 ± 2.23 (172 ± 2.20)	203 ± 4.94 (214 ± 4.00)	102 ± 1.96* (110 ± 1.50)	
120 days	0.11 ± 0.04 (0.13 ± 0.01)	1.52 ± 0.10 (1.48 ± 0.21)	543 ± 5.01** (577 ± 1.80)	150 ± 1.51*** (178 ± 2.00)	192 ± 4.01** (216 ± 2.32)	97 ± 1.01*** (114 ± 1.12)	
180 days	0.10 ± 0.06 (0.12 ± 0.03)	1.40 ± 0.04 (1.50 ± 0.20)	532 ± 4.36** (572 ± 1.64)	142 ± 2.29*** (172 ± 1.00)	185 ± 2.39** (210 ± 2.42)	97 ± 10.14 (121 ± 2.32)	
240 days	0.12 ± 0.05 (0.12 ± 0.02)	1.44 ± 0.03 (1.48 ± 0.08)	531 ± 4.13*** (575 ± 2.81)	147 ± 3.11** (175 ± 1.18)	182 ± 3.14** (221 ± 3.30)	91 ± 15.31 (116 ± 6.68)	
300 days	0.09 ± 0.04 (0.12 ± 0.01)	1.39 ± 0.10 (1.52 ± 0.24)	523 ± 3.34*** (582 ± 1.00)	148 ± 2.61*** (180 ± 1.22)	179 ± 10.50* (220 ± 2.28)	98 ± 2.37*** (120 ± 2.18)	
360 days	0.12 ± 0.09 (0.14 ± 0.01)	1.47 ± 0.011.51 ± 0.24)	520 ± 5.57*** (580 ± 2.24)	145 ± 3.91** (178 ± 1.00)	183 ± 2.61*** (222 ± 1.00)	96 ± 8.31 (118 ± 6.42)	
Treatment withdrawal							
30 days	0.11 ± 0.04 (0.14 ± 0.02)	1.40 ± 0.06 (1.54 ± 0.22)	468 ± 13.20** (580 ± 3.12)	168 ± 10.28 (168 ± 2.33)	187 ± 11.00* (220 ± 4.64)	107 ± 4.36 (120 ± 3.32)	
90 days	0.12 ± 0.05 (0.13 ± 0.03)	1.46 ± 0.02 (1.55 ± 0.40)	476 ± 10.01*** (577 ± 1.22)	165 ± 8.34 (178 ± 1.20)	189 ± 11.35 (221 ± 4.21)	101 ± 5.31* (126 ± 1.22)	

Control values are given in parenthesis.

* $p < 0.05$.

** $p < 0.01$.

*** $p < 0.001$.

Table 1B

The body weight and organs weight in rats treated orally with the isolated compound ECP I of the ethyl acetate sub- fraction of the benzene chromatographic fraction of the chloroform extract of the seeds of *Carica papaya* at 50 mg/kg b.w./day (values are S.E.M. of three animals per group)

Treatment schedule	Body weight (g)	Brain (g)	Thyroid (g)	Heart (g)	Lungs (g)	Liver (g)	Spleen (g)
Treatment							
60 days	167 ± 1.44 (170 ± 1.84)	1.61 ± 0.11 (1.65 ± 0.12)	0.03 ± 0.01 (0.03 ± 0.02)	0.55 ± 0.19 (0.59 ± 0.16)	1.25 ± 0.07 (1.27 ± 0.22)	3.56 ± 0.04 (3.70 ± 0.42)	0.50 ± 0.09 (0.52 ± 0.16)
120 days	165 ± 3.15* (178 ± 1.32)	1.60 ± 0.21 (1.66 ± 0.21)	0.02 ± 0.01 (0.02 ± 0.01)	0.63 ± 0.05 (0.63 ± 0.21)	1.28 ± 0.31 (1.30 ± 0.42)	3.61 ± 0.35 (3.65 ± 0.32)	0.50 ± 0.07 (0.50 ± 0.12)
180 days	170 ± 1.34* (179 ± 1.54)	1.61 ± 0.03 (1.69 ± 0.28)	0.04 ± 0.02 (0.03 ± 0.01)	0.56 ± 0.07 (0.60 ± 0.14)	1.21 ± 0.21 (1.26 ± 0.21)	3.55 ± 0.30 (3.69 ± 0.42)	0.47 ± 0.21 (0.56 ± 0.20)
240 days	163 ± 6.31 (176 ± 1.32)	1.53 ± 0.11 (1.73 ± 0.24)	0.03 ± 0.02 (0.02 ± 0.01)	0.61 ± 0.05 (0.67 ± 0.24)	1.23 ± 0.29 (1.31 ± 0.12)	3.59 ± 0.07 (3.68 ± 0.58)	0.51 ± 0.05 (0.55 ± 0.16)
300 days	163 ± 5.61 (177 ± 1.68)	1.60 ± 0.33 (1.72 ± 0.88)	0.03 ± 0.01 (0.04 ± 0.02)	0.59 ± 0.17 (0.66 ± 0.21)	1.22 ± 0.24 (1.33 ± 0.50)	3.62 ± 0.27 (3.68 ± 0.21)	0.58 ± 0.16 (0.58 ± 0.21)
360 days	162 ± 9.14 (181 ± 1.89)	1.56 ± 0.21 (1.71 ± 0.61)	0.03 ± 0.02 (0.03 ± 0.01)	0.63 ± 0.09 (0.67 ± 0.22)	1.22 ± 0.28 (1.30 ± 0.28)	3.51 ± 0.32 (3.72 ± 0.80)	0.51 ± 0.13 (0.60 ± 0.36)
Treatment withdrawal							
30 days	165 ± 3.16*** (201 ± 1.20)	1.57 ± 0.05 (1.71 ± 0.81)	0.03 ± 0.01 (0.02 ± 0.01)	0.62 ± 0.11 (0.66 ± 0.12)	1.28 ± 0.45 (1.28 ± 0.51)	3.56 ± 0.13 (3.71 ± 0.62)	0.47 ± 0.27 (0.59 ± 0.16)
90 days	172 ± 1.14*** (211 ± 1.66)	1.63 ± 0.04 (1.73 ± 0.71)	0.04 ± 0.02 (0.04 ± 0.01)	0.61 ± 0.09 (0.68 ± 0.15)	1.25 ± 0.20 (1.28 ± 0.60)	3.65 ± 0.51 (3.75 ± 0.66)	0.52 ± 0.19 (0.59 ± 0.22)
Treatment schedule	Adrenal (g)	Kidney (g)	Testis (mg/100 g b.w.)	Epididymis (mg/100 g b.w.)	Seminal Vesicle (mg/100 g b.w.)	Ventral prostate (mg/100 g b.w.)	
Treatment							
60 days	0.12 ± 0.03 (0.14 ± 0.02)	1.39 ± 0.15 (1.44 ± 0.08)	557 ± 2.21 (560 ± 1.50)	165 ± 2.31 (167 ± 1.33)	209 ± 4.01 (215 ± 2.21)	103 ± 1.56** (114 ± 1.68)	
120 days	0.13 ± 0.05 (0.13 ± 0.04)	1.51 ± 0.15 (1.46 ± 0.27)	549 ± 3.54* (560 ± 1.32)	154 ± 1.62** (170 ± 1.24)	187 ± 5.03** (218 ± 1.69)	95 ± 1.21** (114 ± 2.68)	
180 days	0.11 ± 0.02 (0.14 ± 0.03)	1.43 ± 0.07 (1.46 ± 0.87)	554 ± 3.13* (565 ± 1.81)	145 ± 2.31** (170 ± 2.40)	181 ± 3.29** (223 ± 4.68)	94 ± 11.24 (119 ± 3.21)	
240 days	0.14 ± 0.07 (0.13 ± 0.03)	1.41 ± 0.07 (1.49 ± 0.64)	521 ± 8.22** (568 ± 4.42)	141 ± 3.98** (170 ± 1.99)	180 ± 3.98** (221 ± 6.21)	92 ± 13.42 (122 ± 5.24)	
300 days	0.10 ± 0.03 (0.14 ± 0.02)	1.40 ± 0.11 (1.54 ± 0.72)	537 ± 6.91** (571 ± 2.24)	146 ± 3.52** (175 ± 1.13)	175 ± 10.91* (220 ± 8.30)	97 ± 1.94** (125 ± 2.81)	
360 days	0.13 ± 0.03 (0.12 ± 0.03)	1.44 ± 0.07 (1.52 ± 0.70)	531 ± 7.43** (570 ± 3.36)	143 ± 4.86** (173 ± 1.36)	181 ± 2.31** (224 ± 4.46)	98 ± 5.37* (124 ± 1.84)	
Treatment withdrawal							
30 days	0.12 ± 0.05 (0.13 ± 0.03)	1.42 ± 0.03 (1.54 ± 0.27)	523 ± 9.21** (572 ± 1.48)	158 ± 1.23* (175 ± 5.62)	185 ± 11.41* (221 ± 6.30)	108 ± 3.34 (124 ± 5.80)	
90 days	0.12 ± 0.09 (0.14 ± 0.03)	1.45 ± 0.07 (1.50 ± 0.69)	548 ± 5.41* (570 ± 1.44)	167 ± 7.22 (174 ± 2.36)	187 ± 12.21 (225 ± 8.81)	107 ± 2.62 (125 ± 6.96)	

Control values are given in parenthesis.

* $p < 0.05$.

** $p < 0.01$.

*** $p < 0.001$.

Table 2A

Cauda epididymal sperm characteristics and reproductive tissue biochemical parameters of rats treated orally with the isolated compound MCP I of the methanol sub-fraction of the benzene chromatographic fraction of the chloroform extract of the seeds of *Carica papaya* at 50 mg/kg b.w./day (values are S.E.M. of three animals per group)

Treatment schedule		Sperm characteristics							
		Sperm motility (%)			Sperm viability (%)		Sperm density (million/ml)		Abnormal sperm (%)
Treatment									
60 days		1.0 ± 0.66*** (66 ± 1.24)			34 ± 2.40*** (66 ± 1.80)		15 ± 1.03* (21 ± 1.30)		56 ± 0.85*** (30 ± 1.24)
120 days		Nil (71 ± 2.24)			28 ± 2.00*** (66 ± 1.24)		8 ± 1.00** (18 ± 1.22)		64 ± 1.50*** (35 ± 1.80)
180 days		Nil (65 ± 1.24)			10 ± 1.40*** (60 ± 1.99)		5 ± 1.45*** (20 ± 1.28)		88 ± 1.45*** (37 ± 1.46)
240 days		Nil (61 ± 0.36)			8 ± 1.36*** (64 ± 1.38)		6 ± 0.44*** (19 ± 1.44)		75 ± 1.88*** (33 ± 1.24)
300 days		Nil (60 ± 1.28)			7 ± 1.00*** (59 ± 2.68)		5 ± 1.26*** (22 ± 1.00)		78 ± 1.56*** (38 ± 1.28)
360 days		Nil (63 ± 1.88)			9 ± 0.45*** (63 ± 1.20)		3 ± 1.40*** (23 ± 1.28)		87 ± 1.22*** (32 ± 1.24)
Treatment withdrawal									
30 days		30 ± 2.60*** (64 ± 1.12)			35 ± 2.89** (61 ± 1.24)		11 ± 1.00** (20 ± 1.68)		62 ± 4.41** (30 ± 1.64)
90 days		65 ± 2.22 (63 ± 1.20)			60 ± 2.32 (60 ± 2.24)		19 ± 1.36 (19 ± 1.64)		40 ± 1.42* (29 ± 2.14)
Treatment schedule	Tissue biochemistry								
	Testis			Epididymis			Seminal vesicle	Prostate	
	Cholesterol (mg/g)	Glycogen (mg/g)	LDH (U/mg)	Sialic acid (mg/g)	L-Carnitine (nmol/mg protein)	α-Glucosidase (PNP mg/protein mg)	Fructose (mg/g)	ACP (B.U.)	
Treatment									
60 days	4.28 ± 0.14 (4.32 ± 0.12)	2.13 ± 0.08 (2.11 ± 0.05)	21 ± 0.88 (21 ± 0.26)	5.86 ± 0.02 (5.81 ± 0.12)	1.13 ± 0.01 (1.22 ± 0.21)	96.1 ± 0.21 (96.2 ± 0.32)	5.00 ± 0.02 (5.08 ± 0.55)	2.98 ± 0.05 (3.06 ± 0.08)	
120 days	4.19 ± 0.02 (4.22 ± 0.18)	2.10 ± 0.03 (2.14 ± 0.09)	20 ± 1.00 (22 ± 0.24)	5.77 ± 0.02 (5.70 ± 0.63)	1.18 ± 0.05 (1.20 ± 0.08)	94.5 ± 0.62 (96.7 ± 0.81)	4.91 ± 0.01 (5.00 ± 0.12)	2.78 ± 0.03 (3.12 ± 0.18)	
180 days	4.22 ± 0.14 (4.25 ± 0.36)	2.10 ± 0.07 (2.12 ± 0.03)	20 ± 1.02 (20 ± 0.58)	5.57 ± 0.25 (5.80 ± 0.29)	1.04 ± 0.02 (1.16 ± 0.21)	95.1 ± 0.21 (95.8 ± 0.84)	5.06 ± 0.13 (5.12 ± 0.24)	2.80 ± 0.19 (2.97 ± 0.08)	
240 days	4.23 ± 0.06 (4.23 ± 0.08)	2.07 ± 0.08 (2.09 ± 0.07)	20 ± 1.21 (20 ± 1.20)	5.40 ± 0.07 (5.36 ± 0.62)	1.09 ± 0.02 (1.15 ± 0.14)	94.3 ± 0.21 (96.2 ± 0.76)	4.79 ± 0.27 (4.91 ± 0.54)	2.78 ± 0.15 (2.99 ± 0.21)	
300 days	4.15 ± 0.08 (4.25 ± 0.02)	2.11 ± 0.14 (2.13 ± 0.19)	19 ± 1.41 (20 ± 1.24)	4.79 ± 0.19 (5.68 ± 0.64)	1.07 ± 0.06 (1.11 ± 0.10)	93.1 ± 0.74 (95.1 ± 0.62)	5.04 ± 0.17 (5.00 ± 0.12)	2.87 ± 0.29 (2.90 ± 0.16)	
360 days	4.20 ± 0.02 (4.27 ± 0.28)	2.10 ± 0.05 (2.16 ± 0.03)	21 ± 0.98 (20 ± 0.68)	4.62 ± 0.26 (5.80 ± 0.86)	1.07 ± 0.08 (1.10 ± 0.09)	93.5 ± 0.41 (95.4 ± 0.62)	4.94 ± 0.16 (5.20 ± 0.32)	2.88 ± 0.17 (3.02 ± 0.04)	
Treatment withdrawal									
30 days	4.24 ± 0.34 (4.30 ± 0.64)	2.13 ± 0.06 (2.11 ± 0.08)	19 ± 2.02 (21 ± 0.28)	4.81 ± 0.03 (5.62 ± 0.51)	1.09 ± 0.01 (1.16 ± 0.18)	93.9 ± 0.71 (96.2 ± 0.81)	5.13 ± 0.14 (5.10 ± 0.04)	2.90 ± 0.04 (3.10 ± 0.06)	
90 days	4.25 ± 0.14 (4.26 ± 0.30)	2.11 ± 0.08 (2.13 ± 0.05)	21 ± 0.08 (20 ± 0.39)	4.72 ± 0.05 (5.70 ± 1.26)	1.15 ± 0.01 (1.18 ± 0.21)	93.2 ± 0.36 (95.1 ± 0.93)	4.94 ± 0.08 (5.08 ± 0.08)	2.95 ± 0.05 (3.00 ± 0.02)	

Control values are given in parenthesis.

* $p < 0.05$.

** $p < 0.01$.

*** $p < 0.001$.

Table 2B

Cauda epididymal sperm characteristics and reproductive tissue biochemical parameters of rats treated orally with the isolated compound ECP I of the ethyl acetate sub-fraction of the benzene chromatographic fraction of the chloroform extract of the seeds of *Carica papaya* at 50 mg/kg b.w./day (values are S.E.M. of three animals per group)

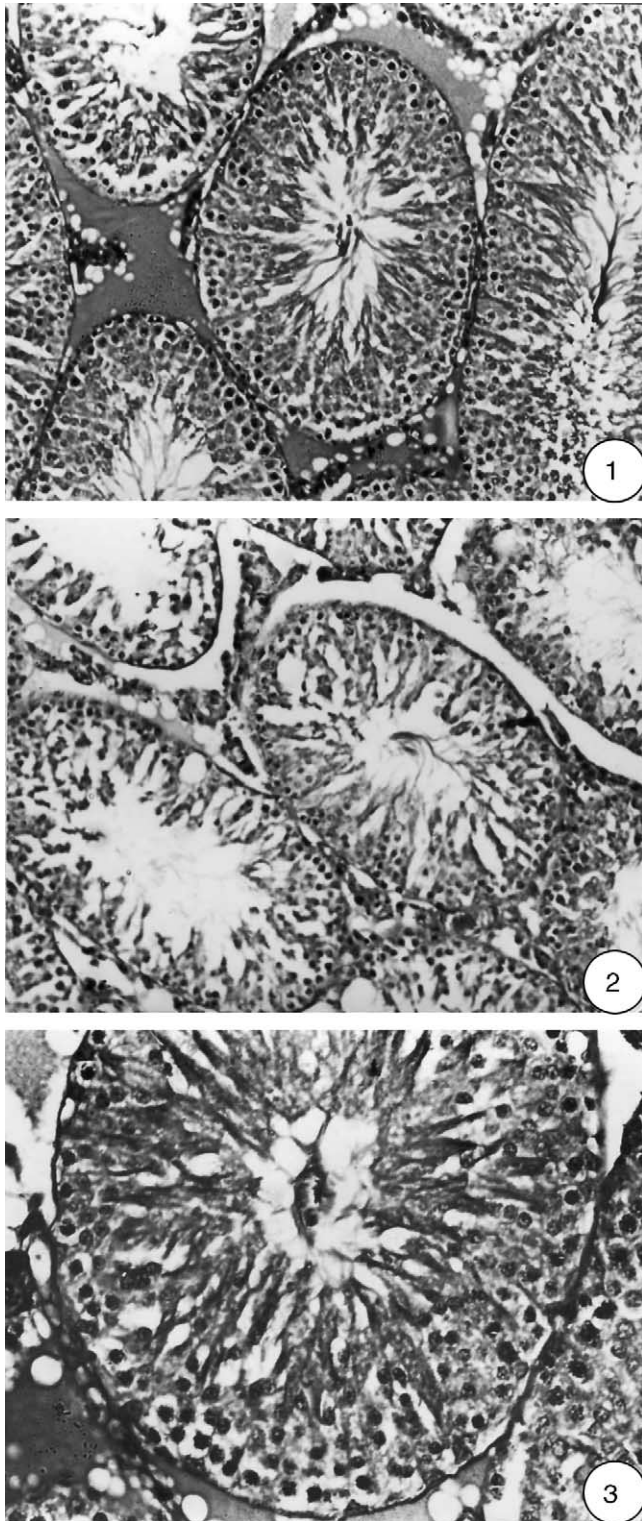
Treatment schedule		Sperm characteristics							
		Sperm motility (%)			Sperm viability (%)		Sperm density (million/ml)		Abnormal sperm (%)
Treatment									
60 days		9 ± 0.21*** (70 ± 1.28)			29 ± 3.14*** (66 ± 1.22)		14 ± 1.23** (23 ± 1.22)		45 ± 1.21** (31 ± 1.20)
120 days		Nil (65 ± 1.04)			23 ± 0.29*** (62 ± 2.28)		13 ± 0.39* (20 ± 1.65)		55 ± 1.15*** (28 ± 2.16)
180 days		Nil (60 ± 1.64)			21 ± 1.41*** (63 ± 0.28)		10 ± 2.31* (21 ± 1.84)		59 ± 1.37*** (29 ± 1.64)
240 days		Nil (67 ± 1.06)			16 ± 1.64*** (63 ± 1.24)		5 ± 1.97** (25 ± 1.62)		71 ± 2.14*** (33 ± 1.80)
300 days		Nil (64 ± 1.21)			15 ± 1.25*** (61 ± 1.08)		4 ± 2.51** (18 ± 1.64)		76 ± 1.25*** (30 ± 1.28)
360 days		Nil (63 ± 1.62)			9 ± 2.91*** (60 ± 2.12)		3 ± 2.39** (22 ± 1.60)		79 ± 1.89*** (35 ± 2.64)
Treatment withdrawal									
30 days		Nil (60 ± 0.98)			7 ± 3.54*** (66 ± 1.26)		9 ± 1.15** (20 ± 1.81)		57 ± 1.23*** (29 ± 1.27)
90 days		42 ± 0.51*** (66 ± 1.36)			56 ± 0.36* (64 ± 1.92)		15 ± 1.09 (21 ± 2.03)		39 ± 0.31*** (26 ± 1.02)
Treatment schedule	Tissue biochemistry								
	Testis			Epididymis			Seminal vesicle	Prostate	
	Cholesterol (mg/g)	Glycogen (mg/g)	LDH (U/mg)	Sialic acid (mg/g)	L-Carnitine (nmol/mg protein)	α-Glucosidase (PNP mg/protein mg)	Fructose (mg/g)	ACP (B.U.)	
Treatment									
60 days	4.43 ± 0.05 (4.40 ± 0.08)	2.13 ± 0.01 (2.27 ± 0.12)	21 ± 0.87 (22 ± 1.42)	5.88 ± 0.05 (6.01 ± 0.24)	1.19 ± 0.03 (1.30 ± 0.27)	96.7 ± 0.34 (98.3 ± 0.66)	5.14 ± 0.02 (5.23 ± 0.08)	2.83 ± 0.07 (3.03 ± 0.09)	
120 days	4.07 ± 0.03 (4.32 ± 0.18)	2.06 ± 0.08 (2.12 ± 0.03)	25 ± 1.25 (21 ± 1.36)	5.84 ± 0.04 (5.78 ± 0.28)	1.18 ± 0.05 (1.22 ± 0.04)	96.5 ± 0.51 (95.0 ± 0.99)	5.08 ± 0.07 (5.02 ± 0.18)	3.01 ± 0.07 (3.12 ± 0.21)	
180 days	4.01 ± 0.07 (4.06 ± 0.02)	1.97 ± 0.08 (2.04 ± 0.12)	20 ± 1.24 (23 ± 1.29)	5.89 ± 0.13 (5.70 ± 0.55)	1.19 ± 0.07 (1.12 ± 0.21)	95.9 ± 0.41 (92.0 ± 1.64)	5.01 ± 0.05 (5.12 ± 0.08)	2.91 ± 0.14 (3.33 ± 0.81)	
240 days	4.27 ± 0.05 (4.03 ± 0.27)	1.76 ± 0.09 (2.05 ± 0.24)	18 ± 1.49 (20 ± 1.60)	5.76 ± 0.21 (5.63 ± 0.28)	1.20 ± 0.10 (1.19 ± 0.08)	96.4 ± 0.11 (96.8 ± 1.26)	5.21 ± 0.16 (5.20 ± 0.15)	3.01 ± 0.09 (3.00 ± 0.12)	
300 days	3.78 ± 0.07 (4.00 ± 0.12)	2.01 ± 0.01 (2.20 ± 0.08)	15 ± 1.69 (20 ± 1.20)	5.38 ± 0.29 (5.80 ± 0.18)	1.21 ± 0.01 (1.28 ± 0.09)	97.1 ± 1.45 (98.2 ± 1.24)	4.94 ± 0.17 (5.25 ± 0.18)	3.02 ± 0.17 (2.96 ± 0.32)	
360 days	3.69 ± 0.18 (4.12 ± 0.08)	1.81 ± 0.04 (2.00 ± 0.21)	15 ± 1.98 (22 ± 1.96)	5.72 ± 0.27 (5.76 ± 0.65)	1.21 ± 0.04 (1.20 ± 0.06)	98.5 ± 2.41 (95.8 ± 2.58)	5.11 ± 0.17 (5.20 ± 0.25)	3.06 ± 0.11 (3.09 ± 0.42)	
Treatment withdrawal									
30 days	3.68 ± 0.03 (4.21 ± 0.62)	2.15 ± 0.09 (2.22 ± 0.27)	21 ± 2.09 (22 ± 1.28)	5.69 ± 0.13 (5.60 ± 0.25)	1.18 ± 0.06 (1.21 ± 0.02)	96.9 ± 0.85 (99.5 ± 1.21)	4.93 ± 0.10 (5.07 ± 0.21)	3.05 ± 0.03 (3.02 ± 0.54)	
90 days	4.28 ± 0.11 (4.33 ± 0.27)	2.14 ± 0.01 (2.12 ± 0.12)	23 ± 0.07 (22 ± 1.20)	5.83 ± 0.06 (5.76 ± 0.28)	1.17 ± 0.08 (1.28 ± 0.08)	97.1 ± 0.37 (97.8 ± 0.69)	5.03 ± 0.06 (5.18 ± 0.08)	3.04 ± 0.02 (3.08 ± 0.02)	

Control values are given in parenthesis.

* $p < 0.05$.

** $p < 0.01$.

*** $p < 0.001$.



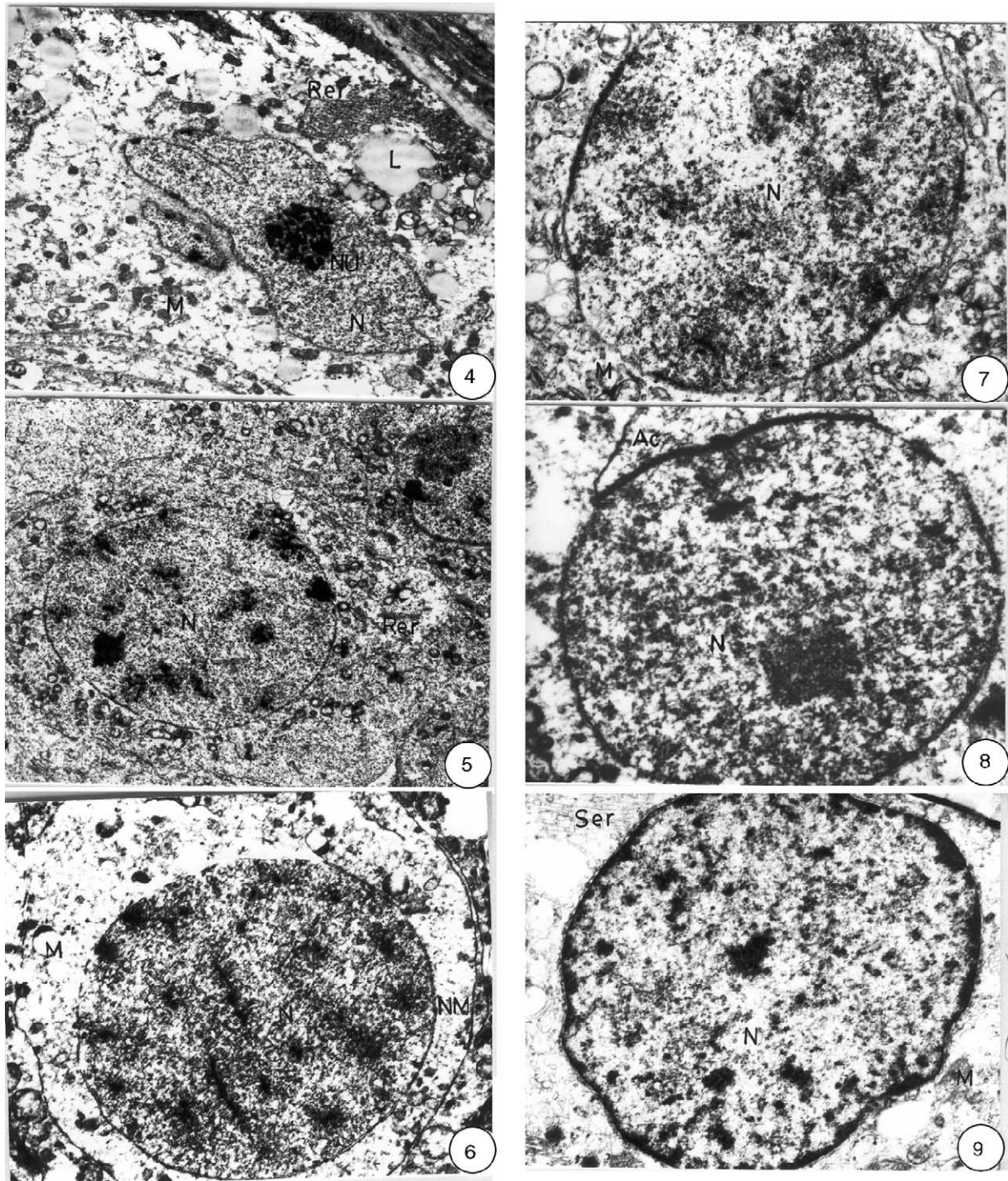
Figs. 1–3. Histology of the rat testis: (1) vehicle control rats depicting normal stages of spermatogenesis, $\times 100$. (2) Testis of rats treated with isolated compound MCP I at 50 mg/kg b.w./day for 240 days. Note the degeneration of seminiferous epithelium, $\times 100$. (3) Testis of rat treated with ECP I. Note the cytoplasmic vacuolization in the Sertoli and germ cells, $\times 400$.

an irregular nucleus with patchy chromatin material, deep indentation and well-defined nucleolus. Perinuclear cytoplasm had cisternae of rough endoplasmic reticulum in close association with lipid droplets. Mitochondria were conspicuous and scattered throughout. Golgi bodies were ill defined. Free ribosomes, glycogen granules and secretory granules were dispersed in the cytoplasm. The plasma membrane was conspicuous, tortuous and showed closer association with adjacent spermatocytes and spermatids. Primary and secondary spermatocytes were rounded and had a prominent nucleus that contained distinct chromatin undergoing condensation, and a well-defined nuclear membrane. The cytoplasm of the primary spermatocyte appeared granular and was characterized by dispersed mitochondria and loose network of rough endoplasmic reticulum. Cytoplasmic organelles of secondary spermatocytes were relatively sparse. Round spermatids were characterized by well-defined nucleus with distinct nuclear membrane and chromatin network and the cytoplasm predominantly occupied with mitochondria. Acrosome formation, marked by acrosomal cap, appeared normal. Leydig cells showed distinct nuclear characteristic, well-defined smooth endoplasmic reticulum and mitochondria surrounding the nucleus (Figs. 4–9).

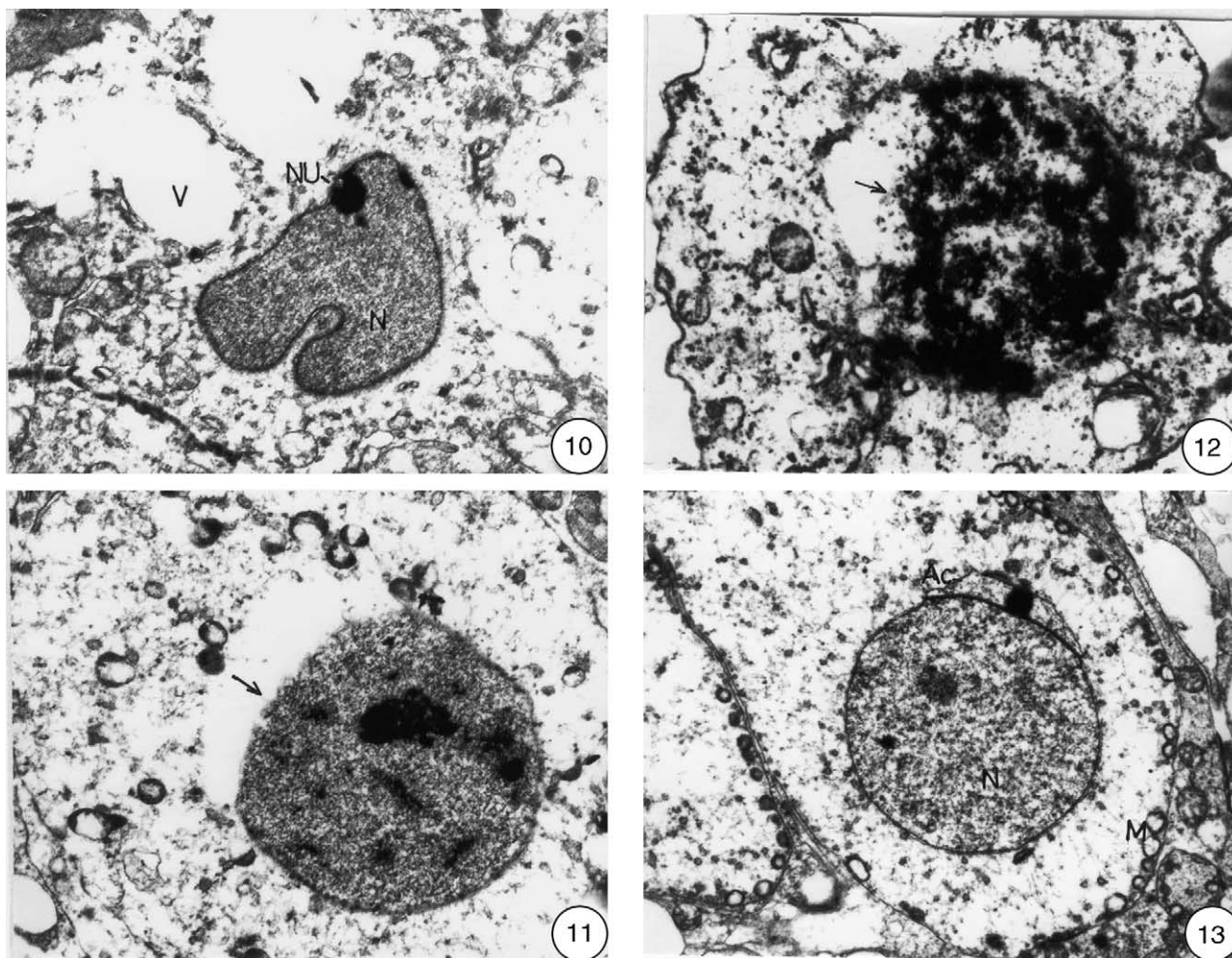
Testicular ultrastructure at 360 days of MCP I treatment reflected altered spermatogenesis. Sertoli cell volume appeared reduced. Although nuclear features appeared normal, the cytoplasm showed vacuolization and sparse organelles. Few mitochondria were scattered in the cytoplasm and showed vacuolization. Nuclear degeneration was evident in the spermatocyte and spermatids, and cytoplasmic organelles in these cells were sparse. Few mitochondria were present in the periphery in spermatids, but these also showed vacuolization. Membrane damage in the acrosomal cap was evident (Figs. 10–13).

3.5.2. Epididymis

Vehicle control rats showed normal ultrastructural features in the epididymal tubules and intertubular elements, particularly in the cauda epididymis where storage and maturation of spermatozoa takes place. The basal (cuboidal) and principal (columnar) cells of the epithelium showed a pseudostratified appearance with apical stereocilia. The nucleus of the former cell type was situated basally and that of the latter cell type more centrally. The lumen contained packed spermatozoa (Fig. 14). In contrast, the epididymis of treated animals showed a reduction in size of the tubules and an increase in the intertubular element; however, the epithelium and the basal and principal cells of the epithelium appeared normal. The lumen contained relatively less spermatozoa, arranged in loose configuration in most of the tubules, and in some cases were devoid of spermatozoa altogether (Figs. 15–16). Principal cells of the cauda epididymis displayed an ultrastructure that reflected active secretory and absorptive functions in the vehicle control (Figs. 17–18) and MCP I exposed (Fig. 19) animals. The normal secretory function was evidenced by the presence of



Figs. 4–9. Ultrastructure of the testis of control rats. *Abbreviations:* L, lipid droplets; M, mitochondria; N, nucleus; NU, nucleolus; NM, nuclear membrane; Rer, rough endoplasmic reticulum; Ac, acrosomal cap; M, mitochondria; N, nucleus; Ser, smooth endoplasmic reticulum. (4) Sertoli cell showing normal cytoplasmic characteristics. The nucleus shows irregular pattern with deep indentation. Single nucleolus and chromatin material are well defined. Cytoplasm shows prominent rough endoplasmic reticulum, numerous mitochondria and lipid droplets, $\times 5000$. (5) The primary spermatocyte depicts prominent nucleus with condensed chromatin structure. Cytoplasm shows prominent mitochondria and loose network of rough endoplasmic reticulum, $\times 5900$. (6) The secondary spermatocyte with prominent nucleus surrounded by well defined double membranous nuclear membrane. Cytoplasmic organelles are relatively sparse. Mitochondria show vacuolisation, $\times 6000$. (7) Round spermatid showing well-defined nucleus. Nuclear membrane is thick and the chromatin material appeared condensed. Cytoplasm showed rich mitochondria, few of them show vacuolisation, $\times 6200$. (8) Spermatid at “cap phase” showing acrosome formation. The membrane of the nucleus as well as acrosome are well defined and the nucleoplasm shows patchy chromatin material, $\times 5900$. (9) Leydig cell shows prominent nucleus with slightly tortuous nuclear membrane and patchy chromatin material. The nucleus surrounded by smooth endoplasmic reticulum; mitochondria appear conspicuous, $\times 5800$.



Figs. 10–13. Ultrastructure of the testis of rats treated with isolated compound MCP I of the methanol sub-fraction of the benzene chromatographic fraction of the chloroform extract of the seeds of *Carica papaya* at 50 mg/kg b.w./day for 360 days. Abbreviations: Ac, acrosomal cap; M, mitochondria; N, nucleus; NU, nucleolus; V, vacuole. (10) Sertoli cell: the cytoplasm shows vacuolization and the cell organelles are relatively sparse. Mitochondria show vacuolization, nucleus show indentation with patchy chromatin and single nucleolus. The volume of the nucleoplasm is relatively less, $\times 5000$. (11) Primary spermatocyte: the nucleus shows degenerative features (arrow). Nuclear membrane is ill defined. Cytoplasmic organelles are sparse. Mitochondria show vacuolisation, $\times 5600$. (12) Round spermatid: The nucleus show degenerative features (arrow), although, contains patchy chromatin material. Nuclear membrane is ill conspicuous. Cell organelles are relatively less and mitochondria show vacuolisation, $\times 5600$. (13) Spermatid at “cap phase”. The membrane of the acrosomal cap is ill defined, peripheral mitochondria showed vacuolization. The cytoplasmic organelles are sparse, $\times 5000$.

numerous well-developed mitochondria in the supra-nuclear region and well-defined Golgi bodies, cisternae of rough endoplasmic reticulum, free ribosomes and secretory granules. The normal absorptive function was reflected in the presence of numerous vesicles in the apical region, multivesicular bodies and pinocytotic vesicles at the base of the cilia.

3.5.3. Seminal vesicle

Following completion of 360 days MCP I or ECP I treatment there were no appreciable changes in the ultrastructure of seminal vesicles. Treated and control animals showed a normal outer muscular region lined by secretory epithelium. The epithelium was tubular, cryptic and thrown into many folds. The cells of the epithelium were columnar with basal nucleus and the cytoplasm appeared granular. The central

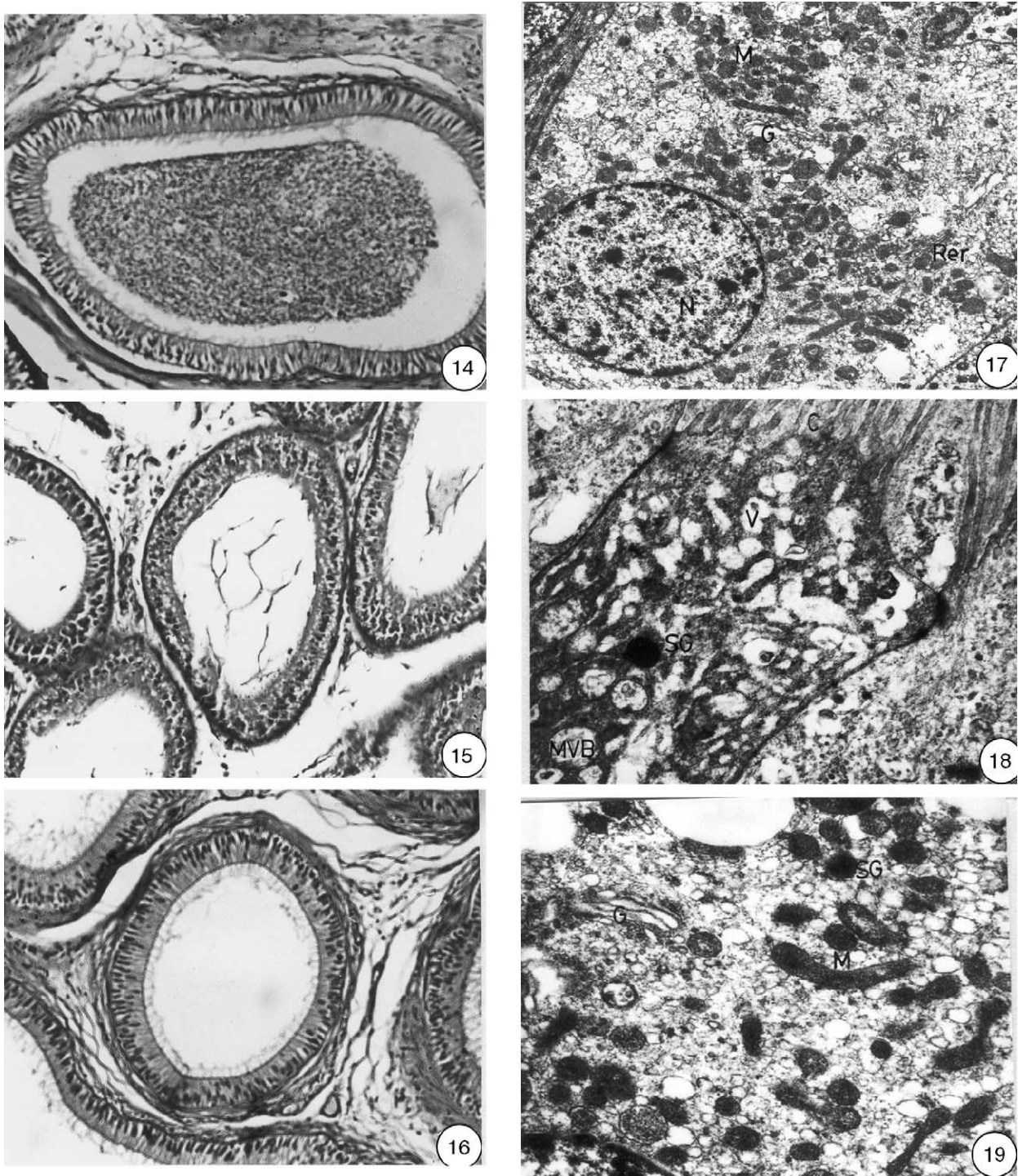
lumen and the cryptic lumen contained eosinated secretory material.

3.5.4. Prostate

Following completion of 360 days MCP I or ECP I treatment there were no appreciable changes in the ultrastructure of the prostate. Treated and control animals showed a normal lobular structure with several prostatic follicles, each lined by columnar epithelium. In the principal cells, the nucleus situated at the central region and the cytoplasm contained eosinated granules. The intertubular region was occupied by connective tissue.

3.6. Fertility and libido

The libido, as observed from pre-coital sexual and mounting behavior of the control and treated animals, remained un-



Figs. 14–19. Histology (Figs. 14–16) and ultrastructure (Figs. 17–19) of the cauda epididymus. *Abbreviations:* C, cilia; G, golgi bodies; M, mitochondria; MVB, multi vesicular bodies; N, nucleus; Rer, rough endoplasmic reticulum; SG, secretory granules; V, vesicles. (14) Vehicle control rat: the epithelium is well defined with basal and principal cells and stereocilia. The lumen is packed with spermatozoa, $\times 100$. (15) Rat treated with isolated compound MCP I at 50 mg/kg b.w./day. Note phagocytes and debris in the lumen, $\times 100$. (16) Rat treated with isolated compound ECP I at 50 mg/kg b.w./day. Note the increase in intertubular elements and lumen devoid of spermatozoa, $\times 100$. (17) Principal cells of the cauda epididymis of vehicle control rat. The nucleus is round with conspicuous chromatin network and situated at the base. The cytoplasm shows numerous well defined mitochondria, prominent Golgi bodies and cisternae of rough endoplasmic reticulum. Secretory granules are scattered, $\times 3600$. (18) Principal cells of the cauda epididymis of vehicle control rat, showing the apical region, characterized by the presence of vesicles and multi vesicular bodies. Cilia are well defined, $\times 2800$. (19) Principal cells of cauda epididymis of rat treated with MCP I at 50 mg/kg b.w./day for 360 days. The cytoplasm depicts active secretory function, indicated by well-defined mitochondria, Golgi bodies, vesicles and secretory granules, $\times 6200$.

Table 3A

Fertility index of rats treated orally with the isolated compound MCP I of the methanol sub-fraction of the benzene chromatographic fraction of the chloroform extract of the seeds of *Carica papaya* at 50 mg/kg b.w./day

Treatment schedule	Animals	Mating ratio*	No. of females delivered	No. of litters	Fertility (%)
Treatment					
60 days	Control	3 males/6 females	6	41	100
	Treated	3 males/6 females	0	0	0
120 days	Control	3 males/6 females	6	32	100
	Treated	3 males/6 females	0	0	0
180 days	Control	3 males/6 females	6	38	100
	Treated	3 males/6 females	0	0	0
240 days	Control	3 males/6 females	6	35	100
	Treated	3 males/6 females	0	0	0
300 days	Control	3 males/6 females	6	40	100
	Treated	3 males/6 females	0	0	0
360 days	Control	3 males/6 females	6	34	100
	Treated	3 males/6 females	0	0	0
Treatment withdrawal					
30 days	Control	3 males/6 females	6	35	100
	Treated	3 males/6 females	0	0	0
90 days	Control	3 males/6 females	6	33	100
	Treated	3 males/6 females	4	26	66.6

* Mating ratio 1 male:2 females.

affected throughout the study period. Monthly fertility tests (shown at 60-day intervals in tables) culminated in normal pregnancy outcome in control animals without significant mortality and morbidity. In the MCP I and ECP I treated animals, monthly fertility tests, beginning at 30 days of treatment, indicated anti-fertility effects until 360 days treatment and 30 days recovery. Treatment withdrawal for 90 days led

to partial recovery of fertility and normal pregnancy outcome (Tables 3A and 3B).

4. Discussion

Bioactive substances extracted from the organic phase of *C. papaya* seeds, when chronically administered to rats by

Table 3B

Fertility index of rats treated orally with the isolated compound ECP I of the ethyl acetate sub-fraction of the benzene chromatographic fraction of the chloroform extract of the seeds of *Carica papaya* at 50 mg/kg b.w./day

Treatment schedule	Animals	Mating ratio*	No. of females delivered	No. of litters	Fertility %
Treatment					
60 days	Control	3 males/6 females	6	33	100
	Treated	3 males/6 females	0	0	0
120 days	Control	3 males/6 females	6	38	100
	Treated	3 males/6 females	0	0	0
180 days	Control	3 males/6 females	6	30	100
	Treated	3 males/6 females	0	0	0
240 days	Control	3 males/6 females	6	34	100
	Treated	3 males/6 females	0	0	0
300 days	Control	3 males/6 females	6	36	100
	Treated	3 males/6 females	0	0	0
360 days	Control	3 males/6 females	6	31	100
	Treated	3 males/6 females	0	0	0
Treatment withdrawal					
30 days	Control	3 males/6 females	6	33	100
	Treated	3 males/6 females	0	0	0
90 days	Control	3 males/6 females	6	38	100
	Treated	3 males/6 females	3	13	50

* Mating ratio 1 male:2 females.

daily oral exposure, led to reduced sperm counts and total inhibition of sperm motility within 90 days of treatment. The present study has extended earlier findings on the crude and partially purified seed extracts by testing the bioactivity isolated from at least two major components of the methanol and ethyl acetate subfractions of the benzene chromatographic fraction of the chloroform extract [1,5,8]. In all these investigations the effects were free of general toxicity as reflected in the weight of vital organs, hematology and serum clinical biochemistry, and free of hormonal mediation inferred through unaltered serum testosterone level [8].

The present study does not reveal the mechanism(s) by which the bioactive compounds in MCP I and ECP I alter sperm motility. Since spermatozoa are matured and stored in the epididymis, any contraceptive agent that could inhibit sperm motility might do so indirectly through disruption of epididymal epithelial cell function (or directly by affecting sperm enzymes) [25]. We detected no significant alterations in the epididymal epithelial ultrastructure, which presented normal protein synthetic machinery as evidenced by well-defined Golgi bodies, rough endoplasmic reticulum, free ribosomes and secretory granules. Neither was there a demonstrable effect on the principal markers of epididymal function, including sialic acid, L-carnitine, α -glucosidase levels in the epididymis. Therefore, these preliminary findings argue against an indirect mechanism of MCP I and ECP I through the epididymal milieu and lead us to speculate a direct mechanism on sperm.

Several possible mechanisms could account for the potential direct effects on spermatozoa. It has been reported that an excessive generation of reactive oxygen species (ROS) might lead to a loss of sperm function and motility [26–27]. High ROS production leads to oxidative damage of cellular lipids, proteins and DNA that can correlate with loss of sperm function [28–29]. Intracellular Ca^{2+} is also known to play a major role in sperm motility [30], and inhibition of intracellular Ca^{2+} in the spermatozoa of rats treated with triptolide, the glycosides of *T. wilfordii* [31] and in the triptolide treated human spermatozoa in vitro [3], resulted in asthenozoospermia. Glycolytic ATP production may be a target, because dyenin ATPase is necessary for normal motility; a positive correlation has been indicated between the motility pattern and sperm glycolysis in the ornidazole treated male rats [32].

Since the present findings indicate a decrease in sperm count in MCP I and ECP I treated rats after 30 days of treatment, a testicular effect of exposure cannot be ruled out. It is well known that Sertoli cells play an important role in germ cell maturation and that these cells are quite susceptible to damage [33]. The presence of agranular and stacks of granular endoplasmic reticulum, numerous mitochondria and multiple Golgi bodies in the basal part of the Sertoli cell cytoplasm are characteristics to Sertoli cells that are metabolically active in protein and steroid biosynthesis [34]. Proteins such as androgen binding protein (ABP) are secreted at maximal rates in the testis during spermatid elongation and spermiation [35], and ABP secreted by Sertoli cells may be required

for specific steps of germ cell maturation [36]. In addition to ABP secretion, the Sertoli cell may capture ABP basally and then transport it to the adluminal aspect and then transfer it to the germ cells [36]. Consistent with reduced protein secretory function, Sertoli cells of a regressed testis do not contain conspicuous stacks of granular endoplasmic reticulum [34]. In the present investigation, treatment with the bioactive compounds in MCP I and ECP I resulted in vacuolization of Sertoli cell cytoplasm and loss of cytoplasmic organelles, suggesting loss of metabolic activity. Thus, degeneration and maturational arrest of germ cells particularly spermatocytes and spermatids observed here could be attributed to the Sertoli cell factors, responsible for germ cell maturation.

Huynh et al. [4] reported in the triptolide treated rat that drug action on spermatozoan morphology could be manifested either directly on maturing germ cells or indirectly on the matured spermatozoa in the epididymis. An important factor in determining sperm head shape is the degree of chromatin packing during spermiogenesis. Head abnormalities detected at short time intervals in epididymal sperm implies damage to sperm nuclei at a late elongated spermatid stage. Persistence of teratozoospermia for longer duration implies damage to Sertoli cells, which then affects germ cells and leads secondarily to a faulty spermatid differentiation [37].

The present study suggests a possible effect of the drug on sperm characteristics mediated through Sertoli cell factors. Histological and ultrastructural evidences supported the notion that long-term exposure to either MCP I or ECP I altered spermatid maturation parameters, particularly nuclear decondensation, acrosome formation and mitochondria, which might result in the production of defective spermatozoa and failure to attain motility characteristics in the cauda epididymis. However, the lack of alterations for a number of biochemical parameters in the epididymal fluid further suggests that the target of drug action on sperm is causally related to the same alterations that affected testicular morphology. Further studies are in progress to address sperm glycolysis and intracellular calcium of spermatozoa and whether or not these factors contribute to the inhibition of sperm motility.

The unaltered body weight and the weight of vital organs, hematology and serum clinical biochemistry of treated rats suggests that the purified products are free of generally anti-metabolic properties. Normal serum testosterone levels and lack of alterations in some androgen sensitive biochemical parameters, particularly fructose of seminal vesicle and ACP of prostate, together with normal morphology of these organs suggest that the adverse effect on sperm is not mediated by general alterations in the endocrine status. Reductions in the weight of the seminal vesicles and prostate could not be related to hormonal modulation.

It is concluded that the compounds MCP I and ECP I are equally effective in terms of male contraception and that the effects could be mediated through testis without untoward side effects. Although the altered parameters did not recover to control levels, following 90 days of treatment withdrawal, the results shown here indicate that partial recovery is

possible following treatment withdrawal, in support of male contraceptive efficacy.

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